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Organic and Inorganic Geochemical Heterogeneity and Cyclicity in Organic-Rich Carbonate Source Rocks

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Abstract

Organic-rich source rocks play a vital role in hydrocarbon exploration and serve as valuable records of past environmental conditions. This study investigates the Upper Cretaceous (Maastrichtian) Type II-S marine source rocks of Jordan (JSR) to examine organic and inorganic compositional variability corresponding to paleoenvironmental conditions during deposition. The identification of distinct geochemical cycles in these source rocks contributes to a deeper understanding of the depositional settings and paleoenvironmental dynamics influencing organic matter distribution.

The study utilized a comprehensive analytical approach, including Spectral Gamma Ray (SGR), RockEval pyrolysis, XRD, XRF, ICP, SEM-EDX, and petrographic examination. These techniques were employed to evaluate the organic and inorganic geochemical composition and distribution of organic matter at both micro and nanoscale levels. A research well drilled in the Al-Lajjun Basin of central Jordan provided a pristine JSR core interval for the detailed geochemical analysis. The characterization of the samples was carried out to understand the lithologic and geochemical variation within the rock formation and correlate it with paleoenvironmental changes.

The JSR is an organic-rich, type IIS, carbonate-dominated source rock interval, deposited during a significant nutrient upwelling event in the Cretaceous time period. It consists of three distinct 3rd-order geochemical cycles that represent geochemical variability driven by paleoclimatic changes during deposition. Cycle 1 features organic-rich mudstone facies with high total organic carbon (TOC), high carbonates, and low silicate mineral composition. The organic matter, found in fossil chambers and the matrix, suggests high productivity with periodic euxinic bottom water conditions and minimal detrital input, aiding in excellent preservation. Cycle 2 transitions to silica-rich mudstone to wackestone facies, marked by increased biogenic silica content and high TOC. Enhanced nutrient upwelling during this time led to higher organic matter productivity and euxinic bottom water conditions, leading to significant organic matter

preservation. Cycle 3, comprising foraminifera-rich wackestone to packstone facies, shows a mix of detrital and biogenic input, with organic matter mainly in the matrix. Despite lower productivity and preservation than the earlier cycles, conditions remained favorable for organic matter accumulation. The intercalation of these cycles indicates the changing environmental and ocean current dynamics at the southern margin of the Tethys Ocean, resulting in mineralogical variations, bottom water redox conditions, and differential dilution, affecting the source rock quality and organic matter preservation in these cycles.

The identified geochemical cycles emphasize the dynamic paleoenvironmental conditions influencing both organic and inorganic geochemical composition within the JSR. Understanding this geochemical heterogeneity is essential for comprehending the depositional architecture and variable paleoenvironmental conditions during source rock deposition. These findings provide a strong foundation for lateral correlation and the characterization of geochemical heterogeneity and cyclicity on a regional scale.

Introduction

Organic-rich mudrocks serve as essential archives of paleoenvironmental conditions and are critical to understanding the factors controlling the accumulation and preservation of organic matter (OM). These source rocks typically develop in depositional settings characterized by high biological productivity, driven by nutrient-rich upwelling, favorable climatic conditions, and oceanic circulation, alongside bottom-water anoxia, which limits OM degradation [1]. While marine environments are commonly associated with OM preservation, non-marine settings such as lakes and swamps also support accumulation due to seasonal stratification, water saturation, and rapid burial [2, 3]. Furthermore, interactions between OM and mineral matrices, particularly clays and carbonates, enhance stabilization by reducing microbial accessibility [4]. Through detailed organic and inorganic geochemical analyses, these depositional and diagenetic conditions can be reconstructed [5-9].

The Cretaceous period, particularly the Late Cretaceous, witnessed widespread deposition of organic-rich sediments driven by high global sea levels, expanded shelf areas, oceanic anoxic events (OAEs), and enhanced productivity [10]. One notable example is the Maastrichtian-aged Jordan Source Rocks (JSR), also known as the Jordan Oil Shales (JOS), found within the lower Muwaqqar Chalk Marl (MCM) Formation. These rocks represent one of the most organically enriched sequences globally, with total organic carbon (TOC) values reaching up to 25 wt.% in some intervals [11, 12]. Despite their richness, the JSR remains thermally immature and has not yet yielded hydrocarbons at a commercial scale [13].

Geologically, Jordan lies on the northwestern margin of the Arabian Plate, adjacent to the Dead Sea Transform Fault system (Fig. 1). During the Late Cretaceous–Eocene, the region formed part of a shallow marine shelf along the southern Neo-Tethys Ocean [14]. The stratigraphy is dominated by the Ajlun and Belqa Groups [14]. The JSR interval is hosted in the lower part of the Maastrichtian MCM Formation, characterized by organic-rich carbonate mudstones deposited under reducing conditions during episodes of elevated marine productivity [15, 16]. The JSR extends across central Jordan and occurs both at the surface and in the shallow subsurface, underlying nearly 70% of the country [17]. This broad distribution, coupled with exceptional organic richness, has made the JSR a focal point of studies exploring its shale oil potential [18-20]. Equivalent Maastrichtian–Paleocene sequences are recognized throughout the Eastern Mediterranean region, associated with a regional upwelling system along the southern Neo-Tethys margin [21-24]. Despite variations in maturity and richness, these strata share geochemical similarities and hydrocarbon potential [6, 22, 25, 26]. Large-scale tectonic and eustatic processes have introduced stratigraphic cyclicity and compositional variability into these successions [27-30]. However, the internal heterogeneities and the key controls on vertical and lateral changes in source rock quality, particularly in the JSR, are still poorly constrained.

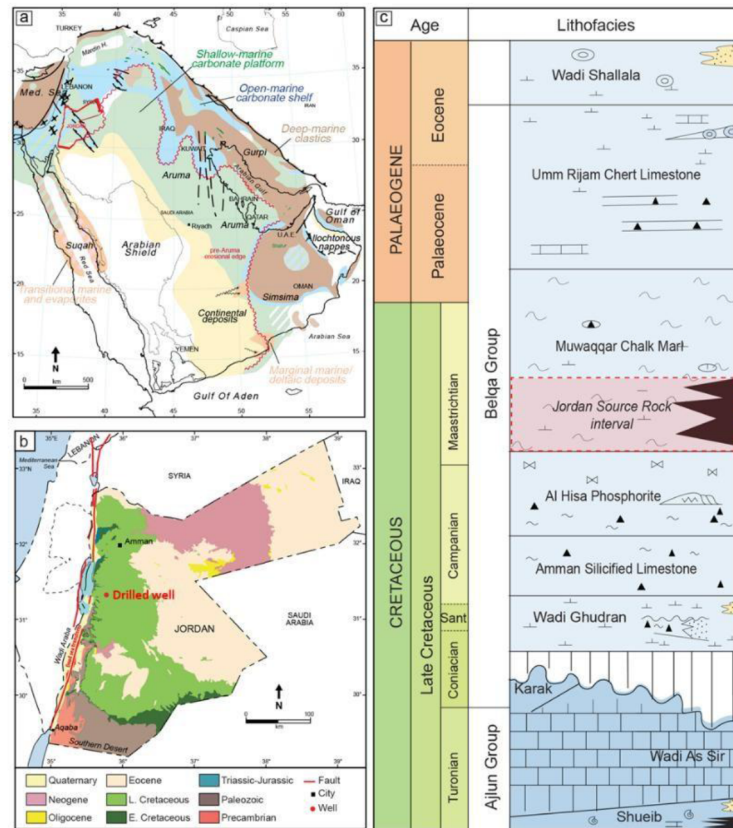


Figure 1—Geological maps and stratigraphy illustrating the regional and local geological context: (a) Depositional facies of the Arabian Plate during the Late Cretaceous [31], (b) Present-day geological map of Jordan showing the location of the studied well, (c) Lithostratigraphic column highlighting the JSR interval within the MCM Formation in the Al-Lajjun Basin, central Jordan [14]

This study integrates high-resolution organic and inorganic geochemical data with petrographic analyses from a fully cored JSR succession. The aim is to unravel the geochemical cyclicity, compositional variability, and depositional controls within the JSR interval. These insights refine our understanding of OM preservation mechanisms and the environmental dynamics that shaped one of the richest source rock systems globally. Beyond the context of Jordan, the findings support improved models of source rock heterogeneity, hydrocarbon generation, and regional stratigraphic correlation across the Eastern Mediterranean.

Methodology

To examine the geochemical composition and heterogeneity of the JSR, a vertical well was drilled to a depth of 72 meters in the Al-Lajjun region of central Jordan (Figure 1b). The top 24 meters represent cuttings, while the remaining 48 meters represent the cored interval. Multiple analytical approaches were utilized to characterize the organic and inorganic components of the JSR interval. These methods included Spectral Gamma Ray (SGR), Rock-Eval pyrolysis, X-ray diffraction (XRD), X-ray fluorescence (XRF), Inductively Coupled Plasma-Optical Emission Spectroscopy (ICP-OES), thin-section petrography, and Scanning Electron Microscopy with Energy Dispersive X-ray spectroscopy (SEM-EDX).

SGR measurements were conducted on the entire core interval at 0.25-meter spacing using an RS-230 gamma-ray spectrometer. The instrument provided concentrations of uranium (U), thorium (Th), and potassium (K), which were used to calculate the total gamma ray (TGR) signal in API units. Rock-Eval pyrolysis was performed with intervals of 1.5 m in the upper cutting section and 0.5 m in the cored interval using a Rock-Eval 7S instrument. The analysis provided critical data on total organic carbon (TOC), free and generated hydrocarbons (S1, S2), CO₂ (S3), Tmax, hydrogen and oxygen indices (HI, OI), production index

(PI), and total sulfur content. Standard procedure to acquire results is mentioned in Behar et al., (2001) [32] and Carvajal-Ortiz et al., (2021) [33]. Quality control included duplicate and standard samples to ensure data accuracy.

A subset of 28 samples was selected for inorganic geochemical analysis based on their organic content. XRD was conducted using a Bruker D8 Advance system to determine mineralogical composition. Samples were powdered to 1–10 μm and scanned over a 5–70° 2 θ range, with mineral phases identified using the ICDD database. Major and minor elements were quantified using a Bruker S8 TIGER XRF spectrometer after pelletizing the powdered samples. For trace element analysis, samples were digested in appropriate acid mixtures depending on carbonate or silica content and analyzed using ICP-OES (Agilent Technologies) to determine trace metal concentrations.

Petrographic studies included both thin-section and SEM-EDX analysis. Selected plugs were examined using a LEICA DM 750P polarizing light microscope to describe grain types, matrix, laminations, porosity, and diagenetic features. Samples were then classified following published petrographic guidelines. Additional polished and titanium-coated plugs were analyzed using a TESCAN FE-SEM coupled with Oxford EDX at KAUST. This enabled high-resolution imaging of mineral phases and organic matter distribution at micro- to nanoscale. This integrated methodological framework provides a comprehensive understanding of the lithological, geochemical, and microstructural variability of the JSR, offering insights into depositional conditions, organic matter enrichment, and preservation mechanisms.

Results & Discussion

Geochemical and Mineralogical Characterization of JSR

Bulk organic geochemical characterization. The JSR interval (24–63 m) demonstrates exceptional source rock characteristics, with total organic carbon (TOC) contents ranging from 2 to 22 wt.% (average 14 wt.%) and consistently high hydrogen index (HI) values (822–894 mg HC/g TOC, average 843 mg HC/g TOC). These parameters indicate the presence of abundant, oil-prone organic matter with excellent preservation potential. The strong covariance between TOC, S₂, and total sulfur content reveals three distinct geochemical cycles within the JSR (Fig. 2). These cycles, characterized by varying organic richness (TOC: 2 – 22%) and sulfur content (0.5 – 5%), reflect changes in depositional conditions. The JSR is characterized as Type IIS kerogen, due to elevated sulfur enrichment, consistent with anoxic to euxinic depositional conditions. These conditions are well-known to enhance the preservation of labile organic compounds and increase the hydrocarbon generation potential of source rocks [34].

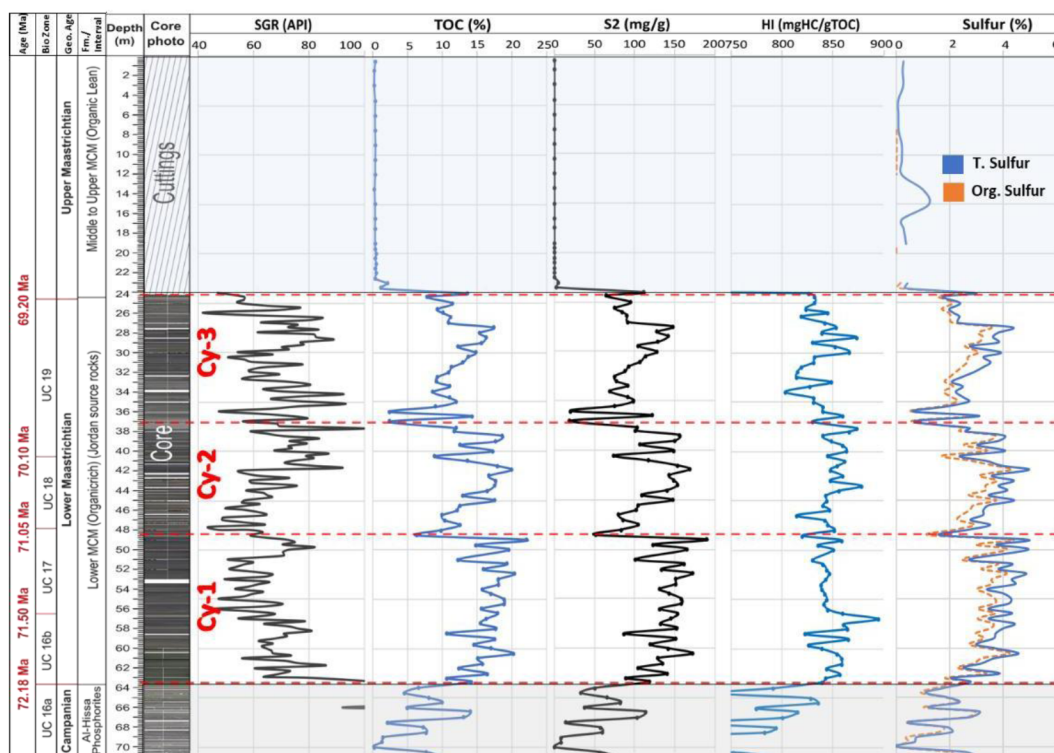


Figure 2—Spectral gamma-ray (SGR) and bulk organic geochemical profiles of the JSR, illustrating the identified geochemical cycles and corresponding biostratigraphic ages within the studied well [36, 37].

Mineralogy (XRD) and Major/Minor Element Composition (XRF). The mineralogical and elemental characteristics of the JSR interval represent distinct depositional patterns that correlate with the previously identified geochemical cycles. XRD analysis demonstrates a predominantly carbonate-rich composition (20-80% calcite), with subordinate silicate minerals (5-56%, primarily quartz and opal-CT) and minor clay content (1-6%, dominated by kaolinite). These mineralogical variations show systematic changes across the three geochemical cycles. Cycles 1 and 3 are characterized by carbonate-dominated assemblages (averaging 70% calcite) with relatively low silicate content (10-20%). In contrast, Cycle 2 exhibits significantly enhanced silica content (6-56%), predominantly as opal-CT derived from biogenic silica, indicating periods of elevated productivity by siliceous microorganisms [35]. XRF analyses corroborate these mineralogical trends, with Ca-dominance reflecting the carbonate-rich nature in cycles 1 and 3, whereas Si-enrichment in Cycle 2 confirms the biogenic silica contribution. These mineral-chemical relationships demonstrate how depositional processes, including variable biogenic productivity and episodic detrital input, controlled both the organic and inorganic components of the JSR, providing a comprehensive framework for understanding compositional variability in organic-rich source rocks.

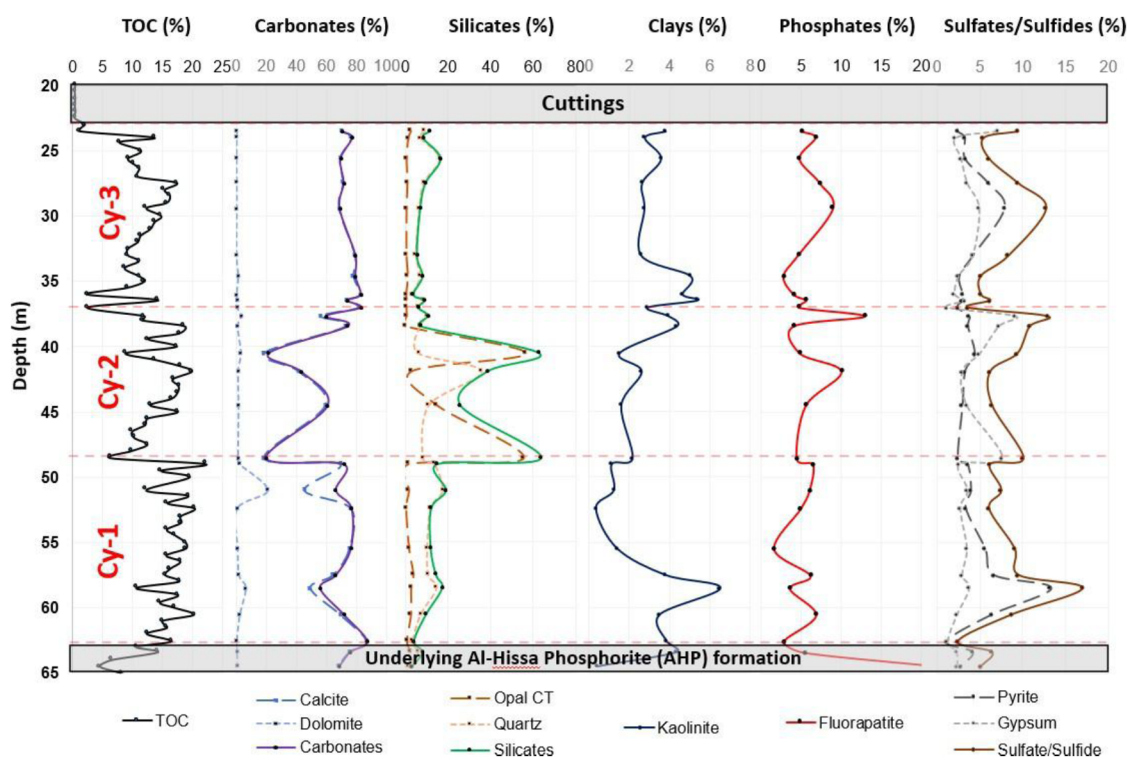


Figure 3—Mineralogical composition of the JSR core section with the identified geochemical cycles. Dotted lines represent individual mineral abundances, while solid lines indicate broader mineral groups [37]

Trace Elemental Characterization and depositional controls. Trace element distributions in the JSR interval provide critical insights into paleoenvironmental conditions during organic matter accumulation. Productivity proxies (Ni, Cu, Zn) show strong correlations with TOC, reflecting enhanced biological productivity during organic-rich intervals [38-40] (Fig. 4). Redox-sensitive elements (U, V, Mo) demonstrate systematic variations, with U and Mo enrichments indicating periodic euxinic conditions [5, 38, 40]. These redox fluctuations likely played a fundamental role in organic matter preservation.

Detrital influences were assessed through Al normalization, revealing two distinct dilution mechanisms: (1) siliciclastic input (evidenced by strong Al correlations) and (2) biogenic silica deposition (indicated by Si excess relative to Al). The JSR shows generally low detrital input (Al = 2-8%), with carbonate dilution dominating in Cycles 1 and 3 (10-20% detrital fraction) and biogenic silica dilution prevailing in Cycle 2. Notably, Cycle 1 exhibits a strong positive Ca-TOC correlation, suggesting coupled carbonate productivity and organic matter accumulation, while Cycles 2 and 3 show more complex relationships where neither detrital nor biogenic dilution alone explains organic matter variability.

The exceptional organic richness and consistently high hydrogen index in the JSR interval provide compelling evidence for sustained high marine productivity. This interpretation is further supported by strong correlations between TOC and productivity-sensitive trace metals (Ni, Cu, Zn), which show concentrations significantly exceeding typical shale values (Fig. 4). The systematic enrichment of these metals, particularly their close association with organic matter, suggests efficient biological uptake by marine algae and subsequent preservation under oxygen-depleted conditions [41, 42].

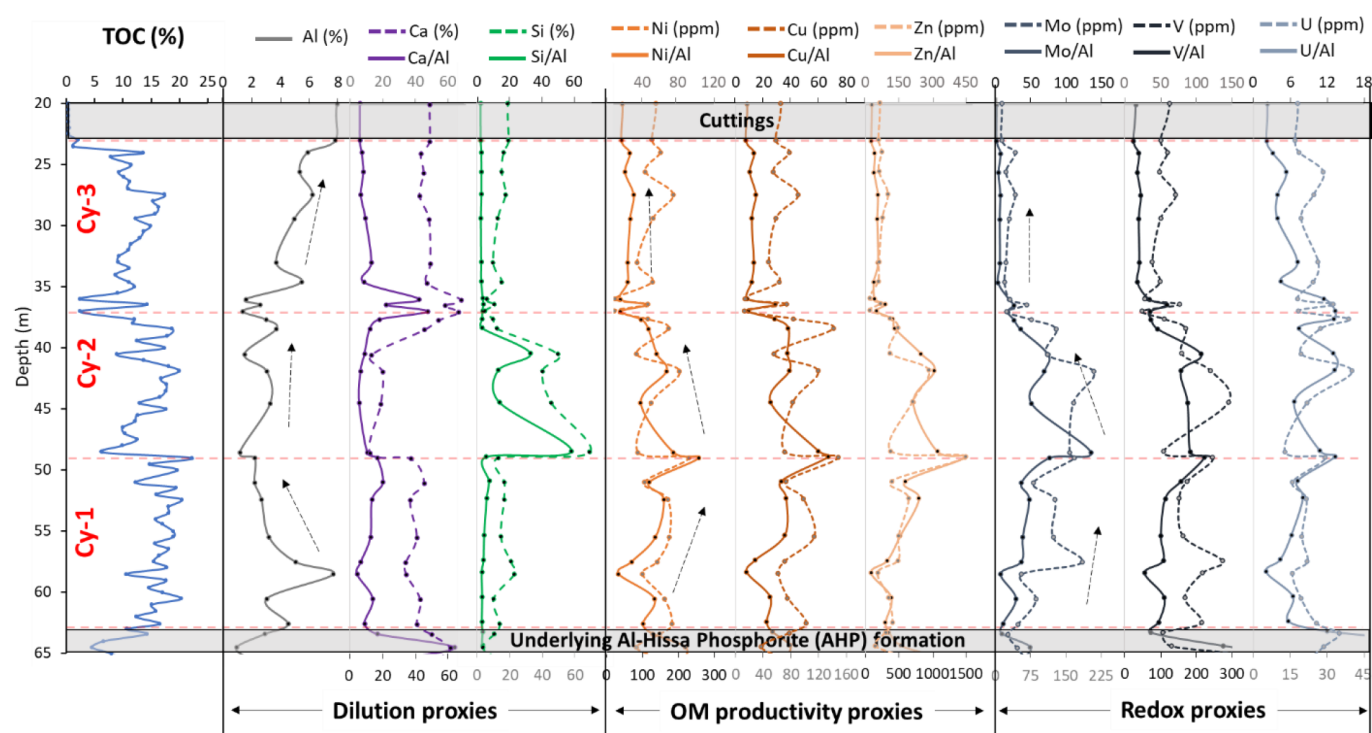


Figure 4—Profiles of elemental concentrations and normalized geochemical proxies for detrital dilution, organic matter productivity, and redox conditions across the JSR cycles. TOC values and average shale composition normalized to aluminum (Al) are shown for comparison [37]

The redox-sensitive trace element distributions (Mo, V, U) in the JSR interval provide critical insights into the evolving oxygen conditions during deposition (Fig. 4). All three elements show significant enrichment, confirming persistently oxygen-depleted bottom waters throughout the studied section. The most reducing conditions occurred during Cycles 1 and 2, where Mo concentrations indicate predominantly euxinic (sulfidic) conditions [37]. The brief excursion at 58.5 m depth in cycle 1, marked by decreased trace metal concentrations, likely represents a short-lived detrital input event rather than a fundamental redox shift. In contrast, Cycle 3 exhibits more moderate Mo enrichment (23–60 ppm), suggesting anoxic to intermittently euxinic conditions. These redox variations correlate strongly with the observed organic matter characteristics, particularly the high sulfur content and Type IIS kerogen composition, demonstrating how sulfate reduction and water column chemistry jointly controlled organic matter preservation. The transition from persistently euxinic (Cycles 1–2) to variable anoxic (Cycle 3) conditions may reflect broader paleoenvironmental changes, including modifications in basin circulation, organic flux, nutrient upwelling conditions, or sea level that affected water mass exchange and ventilation patterns in this Tethyan margin setting [5, 42, 43].

Thin Section Petrography and SEM-EDX Analysis. Thin-section petrography and SEM-EDX analyses reveal significant lithological and compositional variations across the JSR interval. Cycle 1 samples represent organic-rich carbonate mudstones dominated by foraminifera tests and bone fragments in a dark brown matrix, with organic matter preferentially concentrated within fossil chambers (Fig. 5). Cycle 2 comprises silica-rich mudstones- wackestones showing abundant Opal-CT cement and well-preserved foraminifera, while Cycle 3 features foraminiferal wackestone- packstone with lower TOC and matrix-hosted organic matter (Fig. 5).

SEM-EDX analyses highlight these cyclic variations at the micro-scale. Cycle 1 exhibits calcite/ phosphate grains with organic- filled microporosity and framboidal pyrite, indicating sulfate reduction. Cycle 2 shows distinct silica cementation (biogenic Opal-CT) and ZnS precipitates within fossil chambers, confirming biogenic silica origin and euxinic conditions [13, 35] (Fig. 5). In contrast, Cycle 3 displays

cement-filled chambers (calcite/dolomite) and matrix-hosted organic matter, with increased clay content (Fig. 5). These petrographic and microanalytical results demonstrate the varying depositional conditions and diagenetic processes between the JSR cycles, that collectively controlled organic matter distribution and preservation throughout the JSR's depositional history.

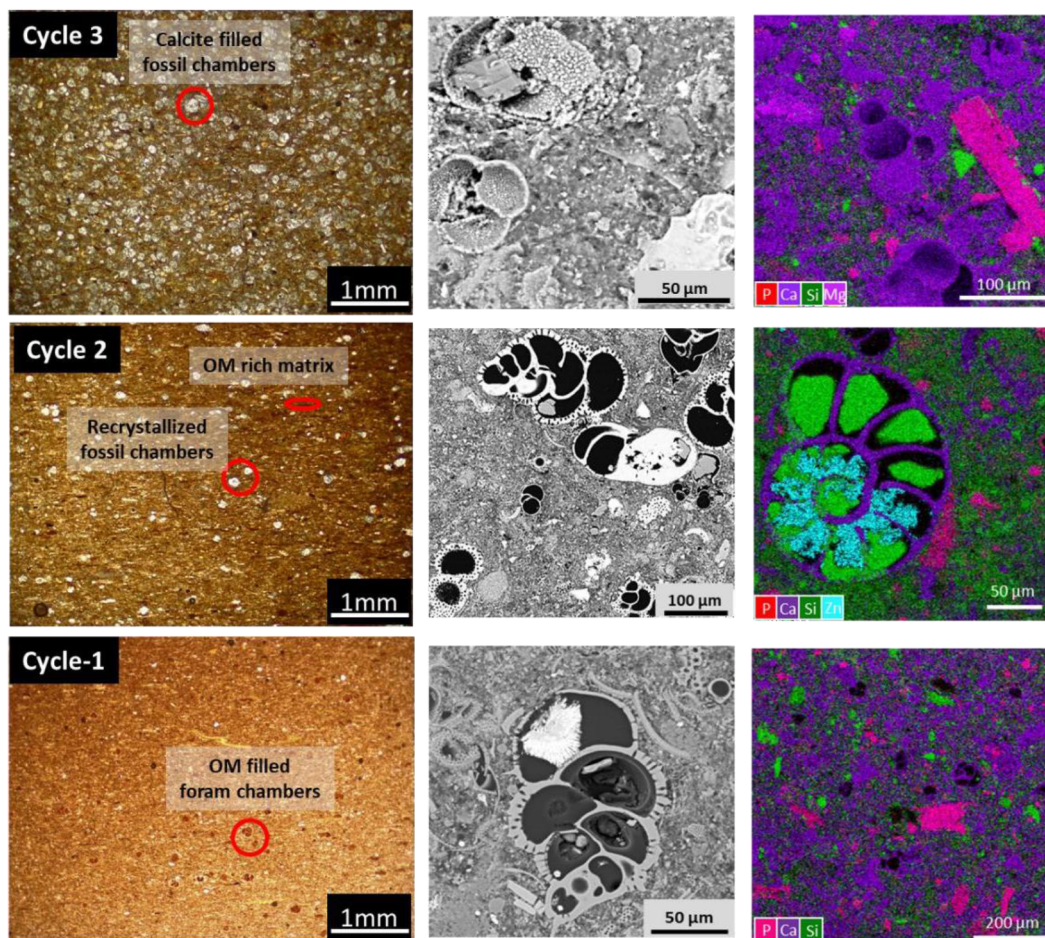


Figure 5—Thin section and SEM-EDX results representing organic matter distribution and elemental compositional variation between the identified cycles [37].

Depositional Cycles: Age, Duration, and Characteristics

The JSR interval records deposition in a dynamic shallow shelf environment along the southern Neo-Tethyan margin, contrasting with traditional models of stagnant pelagic settings. Our integrated data reveal a complex interplay between upwelling-driven productivity, redox fluctuations, and sediment supply that created three distinct third-order depositional cycles (72.2-68.8 Ma) within a broader transgressive phase[36] [37]. These cycles exhibit exceptional organic enrichment (TOC up to 22%) and strong trace metal signatures comparable to global analogues like the Eagle Ford, Posidonia Shale, and La Luna shales.

Cycle 1: Organic-Rich Carbonate Mudstone Facies. Cycle 1, approximately 1.13 Ma in duration and 13.5 m thick, is characterized by high carbonate content and TOC values. Sedimentation rates around 13.3 m/Ma suggest relatively steady accumulation under stable depositional conditions. Bottom waters were predominantly euxinic, promoting exceptional preservation of sulfur-rich Type IIS kerogen (Fig. 4). A distinctive feature of Cycle 1 is the localized detrital event at ~58.5 m depth, marked by a sudden decrease in TOC and a corresponding increase in clay content, likely linked to episodic storm surges, or seismic activity (Fig. 3 & 4). Organic matter and solid bitumen observed within fossil chambers imply early migration of

sulfur-rich bitumen, possibly facilitated by the semi-mobile behavior of sulfur-rich kerogen prior to oil generation [13] (Fig. 5).

Cycle 2: Silica-Rich Mudstone to Wackestone Facies. Cycle 2 spans 1.31 Ma, with a thickness of 11.5 m and a sedimentation rate of 9.2 m/Ma. This cycle exhibits a marked compositional shift to silica-rich sediments, reflecting enhanced production of siliceous biota such as diatoms and sponges. The diagenetic transformation of amorphous biogenic silica (Opal-A) to Opal-CT is well-documented [44, 45]. Elevated concentrations of productivity-related trace metals and paleoredox proxies support the interpretation of increased nutrient upwelling driven by a sustained change in climate and wind patterns over ~1 million years (Fig. 4). The presence of Opal-CT along with organic matter inside the fossil chambers, observed via SEM-EDX, suggests diagenetic processes facilitated simultaneous entrapment of inorganic and organic components, potentially influencing early hydrocarbon migration pathways [46] (Fig. 5).

Cycle 3: Return to Carbonate-Dominated Facies. Cycle 3, lasting 0.91 Ma and 14.5 m thick, marks a return to carbonate dominance, with a reduction in silica content and trace metal concentrations indicative of lower primary productivity and more oxygenated bottom waters (Fig. 3 & 4). Increased aluminum contents point to higher detrital clay influx, possibly from enhanced current activity. Early carbonate cementation is pervasive, filling fossil chambers predominantly with calcite and restricting organic matter occurrence to a dispersed matrix phase (Fig. 5). This cementation likely inhibited early bitumen migration observed in earlier cycles, consistent with a lower sulfur content in organic matter that reduces bitumen mobility.

Paleoenvironmental and Regional Implications

The depositional cycles identified in the JSR correlate well with other late Cretaceous source rocks across the eastern Mediterranean region [22, 23, 26, 30, 47]. The presence of cyclic TOC variations and accompanying mineralogical and trace metal changes provides robust proxies for paleoenvironmental reconstruction. These data support a depositional model of a shallow shelf influenced by nutrient-rich upwelling, modulated by climatic fluctuations and sea-level changes that controlled sedimentation rates, biotic assemblages, and bottom-water redox conditions. The anoxic to euxinic conditions promoted preservation of organic-rich sediments with distinct mineralogical signatures reflecting shifts in dominant biota and sediment supply. High-resolution stratigraphic correlation based on these cycles will offer a powerful tool to refine regional paleoceanographic models and improve predictions of source rock distribution and quality across the MENA region, ultimately informing exploration strategies in analogous basins.

Impact of Compositional Heterogeneity on Unconventional Reservoir Potential

Understanding the spatial and stratigraphic heterogeneity of organic and inorganic geochemical composition is not only critical for assessing petroleum potential but also important for seasonal or permanent gas storage in unconventional reservoirs as well [46, 48-51]. The cyclic alternation between carbonate- and silica-rich facies influences key reservoir properties including mechanical brittleness, thermal maturity, and natural fracture development. Silica-rich intervals cemented by Opal-CT exhibit increased brittleness relative to carbonate-cemented counterparts, enhancing their suitability for hydraulic fracturing stimulation [52]. Additionally, higher thermal conductivity of silica-rich lithologies accelerates hydrocarbon generation and maturation processes, potentially leading to earlier expulsion and more effective migration pathways [46]. Conversely, carbonate-rich intervals with pervasive early cementation may exhibit lower brittleness and reduced organic matter mobility, posing challenges for stimulation and hydrocarbon recovery. The sulfur-rich kerogen within these carbonates, however, may enhance early bitumen migration prior to peak oil generation temperatures, providing localized sweet spots within otherwise less permeable intervals. These insights into compositional heterogeneity enable better targeting of productive zones, refinement of basin models, and optimization of well placement and completion techniques. Recognizing the depositional

cycles and associated geochemical signatures aids in predicting reservoir quality variations and maximizing unconventional resource exploitation.

Conclusion

This comprehensive study of the Maastrichtian MCM Formation in central Jordan integrates spectral gamma ray, bulk geochemical, mineralogical, and petrographic analyses to characterize the depositional environment, hydrocarbon potential, and unconventional reservoir properties of the Jordan Source Rock (JSR) interval. The JSR comprises three distinct third-order depositional cycles, each exhibiting unique geochemical and sedimentological signatures controlled by fluctuations in paleoproductivity, redox conditions, and detrital input.

Cycle 1 represents a carbonate-dominated, organic-rich strata deposited under persistently euxinic conditions, as evidenced by high Mo, V, and U enrichments. The presence of solid bitumen within microfossil chambers suggests early migration of sulfur-rich organics, characteristic of Type IIS kerogen behavior. Cycle 2 marks a shift to silica-rich deposition (Opal-CT up to 56%), driven by intensified upwelling that enhanced siliceous productivity while maintaining euxinic bottom waters. This is also supported by extreme trace metal enrichment and ZnS precipitation in microfossils associated with this cycle. Cycle 3 reflects a return to carbonate dominance with increased detrital input and relatively reduced organic preservation, likely due to enhanced current activity and detrital input.

The cyclic alternation between carbonate- and silica-rich facies highlights the dynamic paleoenvironmental conditions of the southern Neo-Tethyan margin, where sea-level fluctuations and changing ocean circulation patterns modulated nutrient supply and bottom-water redox conditions. The high TOC, Type IIS kerogen, and trace metal signatures correlate well with other Late Cretaceous source rocks in the Eastern Mediterranean, reinforcing a regional model of organic-rich deposition under high-productivity, oxygen-restricted conditions.

The JSR's compositional heterogeneity has critical implications for unconventional reservoir development. Silica-rich intervals exhibit enhanced brittleness, favorable for hydraulic fracturing, while their higher thermal conductivity may promote early hydrocarbon generation. Conversely, carbonate-rich zones, though less brittle, contain mobile sulfur-rich bitumen that could create localized sweet spots. Early diagenetic cementation in Cycle 3 likely inhibited organic matter migration, underscoring the importance of facies-specific reservoir evaluation.

This study establishes the JSR as both a prolific source rock and a potential self-sourcing unconventional play. Future work should integrate basin modeling to refine thermal maturity constraints and experimental studies to assess the mechanical behavior of mixed carbonate-siliceous lithologies during stimulation. These findings provide a robust framework for regional exploration and development of analogous Tethyan margin systems, demonstrating how high-resolution geochemical cyclicity can guide predictions of source rock quality and reservoir performance.

References

1. McCarthy K, Rojas K, Niemann M, Palrnowski D, Peters K, Stankiewicz A. Basic petroleum geochemistry for source rock evaluation. *Oilfield Review* 2011;**23**(2):32-43.
2. Talbot MR, Livingstone DA. Hydrogen index and carbon isotopes of lacustrine organic matter as lake level indicators. *Palaeogeography, Palaeoclimatology, Palaeoecology* 1989;**70**(1):121-37.
3. Meyers PA, Ishiwatari R. The Early Diagenesis of Organic Matter in Lacustrine Sediments. In: Engel MH, Macko SA, editors. *Organic Geochemistry: Principles and Applications*. Boston, MA: Springer US; 1993, p. 185-209.

4. Babakhani P, Dale AW, Woulds C, Moore OW, Xiao K-Q, Curti L, et al Preservation of organic carbon in marine sediments sustained by sorption and transformation processes. *Nature Geoscience* 2025;**18**(1):78-83.
5. Tribouvillard N, Algeo TJ, Lyons T, Riboulleau A. Trace metals as paleoredox and paleoproductivity proxies: An update. *Chemical Geology* 2006;**232**(1-2):12-32.
6. Bou Daher S, Nader FH, Müller C, Littke R. Geochemical and petrographic characterization of Campanian-Lower Maastrichtian calcareous petroleum source rocks of Hasbayya, South Lebanon. *Marine and Petroleum Geology* 2015;**64**:304-23.
7. Sajid Z, Ismail MS, Zakariah MNA, Tsegab H, Vintaned JAG, Hanif T, et al Impact of paleosalinity, paleoredox, paleoproductivity/preservation on the organic matter enrichment in black shales from triassic turbidites of semanggol basin, peninsular malaysia. *Minerals* 2020;**10**(10):1-34.
8. Leushina E, Bulatov T, Kozlova E, Panchenko I, Voropaev A, Karamov T, et al Upper jurassic– lower cretaceous source rocks in the north of western siberia: Comprehensive geochemical characterization and reconstruction of paleo-sedimentation conditions. *Geosciences (Switzerland)* 2021;**11**(8).
9. Fathy D, Baniasad A, Littke R, Sami M. Tracing the geochemical imprints of Maastrichtian black shales in southern Tethys, Egypt: Assessing hydrocarbon source potential and environmental signatures. *International Journal of Coal Geology* 2024;**283**:104457.
10. Jenkyns HC. Cretaceous anoxic events: from continents to oceans. *Journal of the Geological Society* 1980;**137**(2):171-88.
11. Muhaisen Alnawafleh H. Jordanian Oil Shales: Variability, Processing Technologies, and Utilization Options. *Journal of Energy and Natural Resources* 2015;**4**(4):52-.
12. Hakimi MH, Abdullah WH, Alqudah M, Makeen YM, Mustapha KA. Organic geochemical and petrographic characteristics of the oil shales in the Lajjun area, Central Jordan: Origin of organic matter input and preservation conditions. *Fuel* 2016;**181**:34-45.
13. Grohmann S, Littke R, Abu-Mahfouz I, Gaus G, Klaver J, Thüns N, et al The deposition of type II-S Jordan oil shale in the context of Late Cretaceous source rock formation in the Eastern Mediterranean realm. Insights from organic and inorganic geochemistry and petrography. *Marine and Petroleum Geology* 2023;**148**(December 2022):106058-.
14. Powell JH, Moh'd BK. Evolution of Cretaceous to Eocene alluvial and carbonate platform sequences in central and south Jordan. *GeoArabia* 2011;**16**(4):29-82.
15. Abed AM, Amireh BS. Petrography and Geochemistry of Some Jordanian Oil Shales From North Jordan. *Journal of Petroleum Geology* 1983;**5**(3):261-74.
16. Hakimi MH, Abdullah WH, Alqudah M, Makeen YM, Mustapha KA. Reducing marine and warm climate conditions during the Late Cretaceous, and their influence on organic matter enrichment in the oil shale deposits of North Jordan. *International Journal of Coal Geology* 2016;**165**:173-89.
17. Ibrahim KM, Jaber JO. Geochemistry and environmental impacts of retorted oil shale from Jordan. *Environmental Geology* 2007;**52**(5):979-84.
18. Ibrahim KM, Aljurf S, Rahman HA, Gülamber C. Exploration and evaluation of oil shale resources from attarat area, central Jordan. IMCET 2019 - Proceedings of the 26th International Mining Congress and Exhibition of Turkey 2019(February 2020):1116-28.
19. Sakhrieh A, Hamdan M. A STUDY ON THE JORDANIAN OIL SHALE RESOURCES AND UTILIZATION. Global Conference on Power Control and Optimization, Dubai, UAE, 1-3 June 2011 A. 2011.

20. Fathy D, Farouk S, Qteishat A, Ahmad F, Sami M, Al-Kahtany K, et al Geochemical characterization of Upper Cretaceous organic-rich deposits: Insights from the Azraq Basin in Jordan. *Journal of Asian Earth Sciences* 2024;**276**:106365.
21. Abed AM, Arouri KR, Boreham CJ. Source rock potential of the phosphorite-bituminous chalk-marl sequence in Jordan. *Marine and Petroleum Geology* 2005;**22**(3):413-25.
22. Bou Daher S, Nader FH, Strauss H, Littke R. Depositional environment and source-rock characterisation of organic-matter rich upper santonian - upper campanian carbonates, Northern Lebanon. *Journal of Petroleum Geology* 2014;**37**(1):5-24.
23. Yang R, Wang Y, Cao J. Cretaceous source rocks and associated oil and gas resources in the world and China: A review. *Petroleum Science* 2014;**11**(3):331-45.
24. Rosenberg YO, Reznik IJ, Vinegar HJ, Feinstein S, Bartov Y. Comparing natural and artificial thermal maturation of a Type II-S source rock, Late Cretaceous, Israel. *Marine and Petroleum Geology* 2021;**124**(September 2020):104773-.
25. Meilijson A, Ashckenazi-Polivoda S, Ron-Yankovich L, Illner P, Alsenz H, Speijer RP, et al Chronostratigraphy of the Upper Cretaceous high productivity sequence of the southern Tethys, Israel. *Cretaceous Research* 2014;**50**:187-213.
26. Grohmann S, Fietz SW, Littke R, Daher SB, Romero-Sarmiento MF, Nader FH, et al Source rock characterization of mesozoic to cenozoic organic matter rich marls and shales of the Eratosthenes Seamount, Eastern Mediterranean Sea. *Oil and Gas Science and Technology* 2018;73.
27. Vail PR, Audemard F, Bowman SA, Eisner PN, Perez-Cruz C. The stratigraphic signatures of tectonics, eustasy and sedimentology—an overview. *Cycles and Events in Stratigraphy*. 1991, p. 617-59.
28. Ali Hussein M, Alqudah M, Blessenohl M, Podlaha OG, Mutterlose J. Depositional environment of Late Cretaceous to Eocene organic-rich marls from Jordan. *GeoArabia* 2015;**20**(1):191-210.
29. Alqudah M, Ali Hussein M, van den Boorn S, Podlaha OG, Mutterlose J. Biostratigraphy and depositional setting of Maastrichtian - Eocene oil shales from Jordan. *Marine and Petroleum Geology* 2015;**60**:87-104.
30. Fathy D, Wagreich M, Fathi E, Ahmed MS, Leila M, Sami M. Maastrichtian Anoxia and Its Influence on Organic Matter and Trace Metal Patterns in the Southern Tethys Realm of Egypt during Greenhouse Variability. *ACS Omega* 2023;**8**(22):19603-12.
31. Ziegler MA. Late Permian to Holocene paleofacies evolution of the Arabian Plate and its hydrocarbon occurrences. *GeoArabia* 2001;**6**:445-504.
32. Behar F, Beaumont V, De B. Penteado HL. *Rock-Eval 6 Technology: Performances and Developments*. *Oil & Gas Science and Technology* 2001;**56**(2):111-34.
33. Carvajal-Ortiz H, Gentzis T, Ostadhassan M. Sulfur Differentiation in Organic-Rich Shales and Carbonates via Open-System Programmed Pyrolysis and Oxidation: Insights into Fluid Souring and H₂S Production in the Bakken Shale, United States. *Energy and Fuels* 2021;**35**(15):12030-44.
34. Usman M, Israa AM, Volker V, Sebastian G, Ralf L. Unconventional Source Rock Heterogeneities; Effects Of Inorganic Geochemical Composition On Thermal Maturation. International Petroleum Technology Conference (IPTC). 2024:23319-.
35. Huggett J, Hooker JN, Cartwright J. Very early diagenesis in a calcareous, organic-rich mudrock from Jordan. *Arabian Journal of Geosciences* 2017;**10**(12).
36. Messaoud JH, Kalifi A, Alibrahim A, Ibrahim K, Chirakal T, Ardila-Sanchez M, et al Chronostratigraphy of the mixed Upper Cretaceous deposits at the northern margin of the Arabian Plate (Jordan). *Newsletters on Stratigraphy* 2025.

37. Usman M, Ardila-Sanchez M, Idiz E, Abu-Mahfouz IS, van Buchem F, Vahrenkamp V. Organic and inorganic geochemical cyclicity of a Maastrichtian oceanic open-shelf carbonate source rock. *Scientific Reports* 2025;**15**(1):15993.
38. Sageman BB, Murphy AE, Werne JP, Ver Straeten CA, Hollander DJ, Lyons TW. A tale of shales: The relative roles of production, decomposition, and dilution in the accumulation of organic-rich strata, Middle-Upper Devonian, Appalachian basin. *Chemical Geology* 2003;**195**(1-4):229-73.
39. Scholz F. Identifying oxygen minimum zone-type biogeochemical cycling in Earth history using inorganic geochemical proxies. *Earth-Science Reviews* 2018;**184**(May):29-45.
40. Sajid Z, Ismail MS, Tsegab H, Hanif T, Ahmed N. Paleoproductivity / Preservation on the Organic Matter Enrichment in Black Shales from Triassic Turbidites. *Minerals* 2020;**10**(915).
41. Boning P, Brumsack HJ, Böttcher ME, Schnetger B, Kriete C, Kallmeyer J, et al Geochemistry of Peruvian near-surface sediments. *Geochimica et Cosmochimica Acta* 2004;**68**(21):4429-51.
42. Tribouvillard N, Algeo TJ, Baudin F, Riboulleau A. Analysis of marine environmental conditions based on molybdenum- uranium covariation-Applications to Mesozoic paleoceanography. *Chemical Geology* 2012;**324-325**:46-58.
43. Scott M M. Relationships between the trace element composition of sedimentary rocks and upper continental crust. *Geochemistry, Geophysics, Geosystems* 2001;2.
44. Longfei L, Weixin L, Lingjie Y, Wentao Z, Baojian S, Tenger B. *Petroleum Geology & Experiment Early diagenesis characteristics of biogenic opal and its influence on porosity and pore network evolution of siliceous shale*. 2020.
45. Schieber J, Krinsley D, Riciputi L. Diagenetic origin quartz silt in mudstones and implications for silica cycling. *Nature* 2000;**406**(6799):981-5.
46. Usman M, Grohmann S, Abu-Mahfouz IS, Vahrenkamp V, Littke R. Effects of geochemical compositional heterogeneities on hydrocarbon expulsion and thermal maturation: An analog study of Maastrichtian source rocks from Jordan. *International Journal of Coal Geology* 2024;**294**:104587.
47. Marlow BL, Kendall CCG, Yose LA. *Petroleum Systems of the Tethyan Region*. 2014.
48. Usman M, Yekeen N, Ali M, Ali M, Keshavarz A, Iglauer S, et al Influence of organic matter and mineral composition on carbonate source rock wettability: Implications for CO₂ geostorage. *Fuel* 2025;**399**:135666.
49. Ali M, Isah A, Yekeen N, Hassanpouryouzband A, Sarmadivaleh M, Okoroafor ER, et al Recent progress in underground hydrogen storage. *Energy & Environmental Science* 2025.
50. Humphrey JD, Chan SA, Usman M, Albensaad B, Alzayer Y. Geological Characterization of a Late Jurassic Carbonate Mudstone, Saudi Arabia: Sedimentology, Geochemistry, and Geomechanics. AAPG Middle East Region Geoscience Technology Workshop: 3rd Edition Carbonate Reservoirs of the Middle East. **51649**. 2020:6-8.
51. Chan S, Humphrey J, Albensaad B, Usman M, Alramadan K, Alzayer Y. Sedimentology, geochemistry, and mechanical stratigraphy of an Early Kimmeridgian carbonate mudstone in an Arabian intrashelf basin. Unconventional Resources Technology Conference, 20th22 June 2022. Unconventional Resources Technology Conference (URTeC); 2022, p. 1139-46.
52. Iakusheva R, Abu-Mahfouz I, Usman M, Finkbeiner T, Vahrenkamp V. Immature Type II-S Carbonate Source-Rocks in Jordan: Assessing Inherited Source Rock Properties before Maturation and Expulsion. 85th EAGE Annual Conference & Exhibition (including the Workshop Programme). *European Association of Geoscientists & Engineers*; 2024:1-5.